3.8.2 Forming algebraic expressions

We now to translate a few statements into an algebraic language and recall how to frame expressions. Here are some examples:

Statement	Expression	Comment
The sum of 8 and 7	$8 + 7$	When simplified, we get a single number. This is a numerical expression
The sum of x and 7	$x + 7$	We get an algebraic expression $x + 7$, since x is a variable.
16 divided by y	$\frac{16}{y}$	Here, y is a variable.
One more than thrice a number p	$3p+1$	Here p is a variable; 3 is the coefficient of p.
The product of a number and the same number less 3	$x(x - 3)$	Note that here x stands for the same number throughout in the expression. (We use brackets to indicate multiplication).

3.8.3 Equations

An **equation** is a statement that asserts the **equality** of two expressions; the expressions are written one on each side of an "equal to" sign.

For example: $2x + 7 = 17$ is an equation (where x is a variable). $2x + 7$ forms the Left Hand Side (LHS) of the equation and 17 is its Right Hand Side (RHS).

Linear equations

An equation containing only one variable with its highest power as one is called a linear equation. Examples: $3x - 7 = 10$.

Linear equations in one or more variables:

An equation is formed when a statement is put in the form of mathematical terms. Here are some examples:

(i) A number is added to 5 to get 25

This statement can be written as $x + 5 = 25$.

This equation $x + 5 = 25$ is formed by one variable (x) whose highest power is 1. So it is called a linear equation in one variable.

Therefore, an equation containing only one variable with its highest power as one is called a linear equation in one variable.

Examples: $5x - 2 = 8$, $3y + 24 = 0$

This linear equation in one variable is also known as **simple equation.**

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